Exercise No: 5	Remediating an Incident by Using AWS Config and Lambda
Date of Exercise	16.10.2026

AIM:

To implement automated security incident detection and remediation in Amazon Web Services (AWS) using AWS Config and AWS Lambda. To configure IAM roles and permissions, enable AWS Config to monitor EC2 security groups, create a custom AWS Config rule that invokes a Lambda function, and automatically remediate unauthorized inbound rule modifications. To verify the remediation process by analyzing execution logs in Amazon CloudWatch.

DESCRIPTION:

This lab focused on monitoring and automatically remediating security-related changes to Amazon EC2 security groups using AWS Config and AWS Lambda. The permissions assigned to the preconfigured IAM roles were examined, and the required AWS Config managed policy was attached to the appropriate role. AWS Config was then enabled to maintain an inventory of EC2 security group resources within the selected AWS Region.

The activities involved modifying the inbound rules of the LabSG1 security group to simulate a security incident and creating a custom AWS Config rule named EC2SecurityGroup to invoke a preconfigured Lambda function whenever monitored configurations changed. The Lambda function's remediation logic was examined to understand how unwanted inbound rules were identified and removed. Finally, Amazon CloudWatch logs were analysed to verify the automatic removal of the additional SMTPS and IMAPS rules.

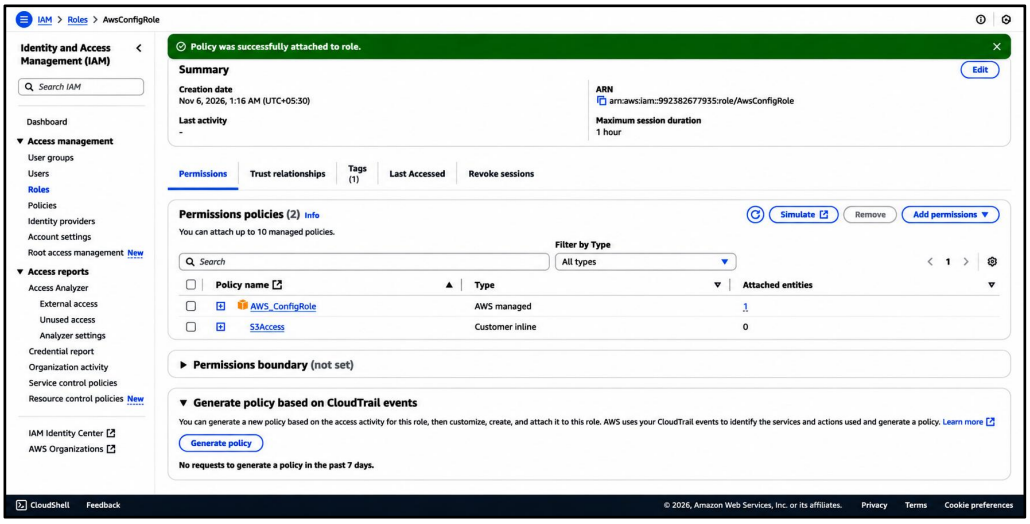
PROCEDURE:

Task 1: Examining and Updating IAM Roles

The IAM console was opened, and the permissions attached to the preconfigured AwsConfigLambdaSGRole were examined. The awsconfig_lambda_ec2_sg_role_policy was reviewed to understand the permissions granted to the Lambda function, including CloudWatch Logs operations, AWS Config compliance evaluation updates, and EC2 security group ingress rule modifications. The AwsConfigRole was then examined, and the AWS_ConfigRole managed policy was attached to provide AWS Config with the required permissions to monitor resources and record configuration changes.

Result: The permissions of both IAM roles were examined and configured to support the interaction between AWS Config, AWS Lambda, and Amazon CloudWatch.

OUTPUT:

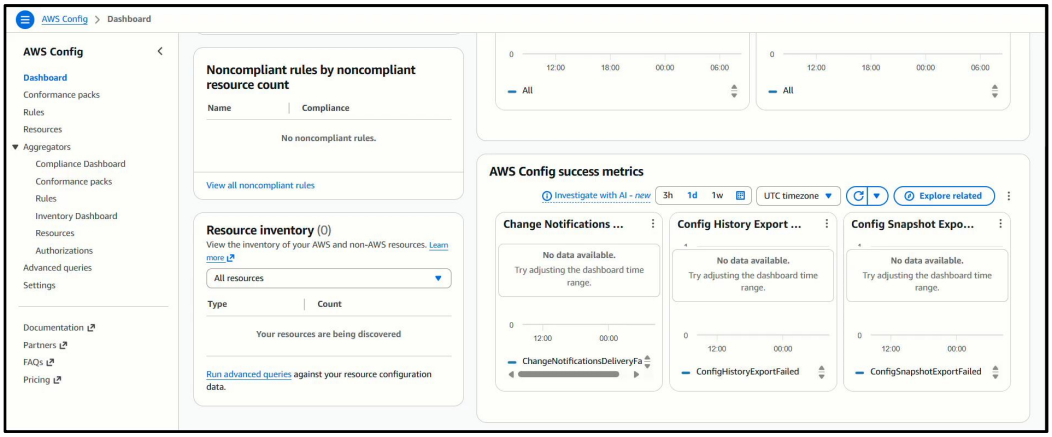


Task 2: Setting Up AWS Config to Monitor Resources

The AWS Config service was opened, and the resource recording settings were configured to monitor specific resource types. The recording strategy was set to Specific resource types, with AWS EC2 SecurityGroup selected and the recording frequency set to Continuous. The existing AwsConfigRole was selected as the IAM role for AWS Config, and the default S3 delivery channel settings were retained. After reviewing the configuration, AWS Config was enabled. The Resource Inventory section was then opened to examine the EC2 security groups and related resources discovered in the selected AWS Region.

Result: AWS Config was successfully configured to continuously monitor EC2 security groups, and the resource inventory was examined to verify the discovery of monitored resources.

OUTPUT:



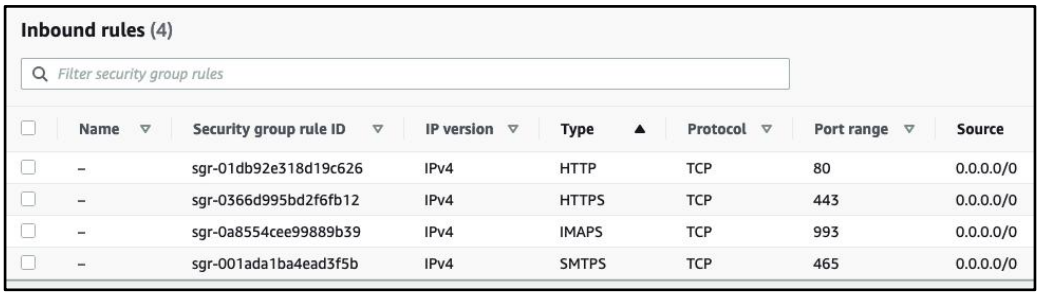
Task 3: Modifying a Security Group That AWS Config Monitors

The Amazon VPC console was opened, and the Lab VPC was selected to locate the LabSG1 security group. The existing HTTP inbound rule was modified to allow traffic from

Anywhere-IPv4. Additional inbound rules were then created for HTTPS, SMTPS, and IMAPS, with Anywhere-IPv4 configured as the source for each rule. After reviewing the inbound rule settings, the changes were saved to simulate a security incident involving unauthorized or noncompliant security group modifications.

Result: The LabSG1 security group was modified to allow inbound HTTP, HTTPS, SMTPS, and IMAPS traffic from Anywhere-IPv4. These changes introduced additional access permissions that would later be detected and remediated by AWS Config and Lambda.

OUTPUT:



Inbound rules (4)

Filter security group rules

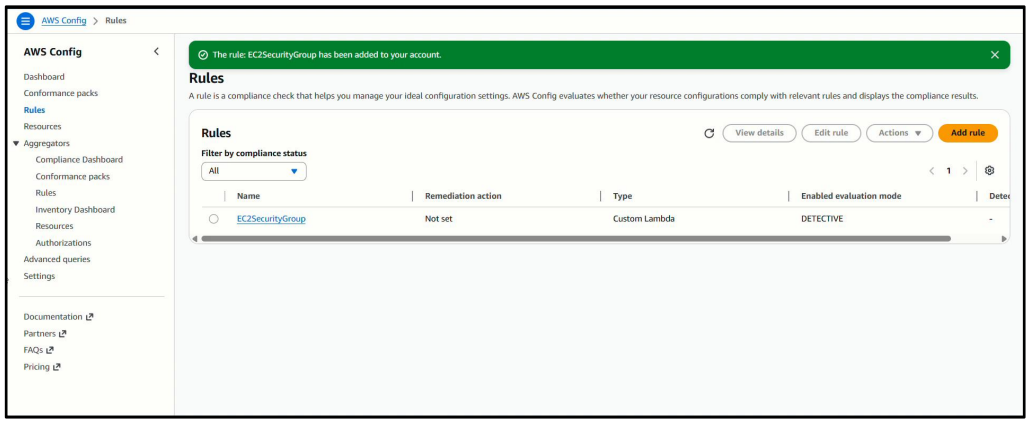
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-01db92e318d19c626	IPv4	HTTP	TCP	80	0.0.0.0/0
–	sgr-0366d995bd2f6fb12	IPv4	HTTPS	TCP	443	0.0.0.0/0
–	sgr-0a8554cee99889b39	IPv4	IMAPS	TCP	993	0.0.0.0/0
–	sgr-001ada1ba4ead3f5b	IPv4	SMTPS	TCP	465	0.0.0.0/0

Task 4: Creating an AWS Config Rule That Calls a Lambda Function

The Lambda function ARN provided in the AWS Details section was copied and used while creating a custom AWS Config rule. In the AWS Config console, the Create custom Lambda rule option was selected. The rule was named EC2SecurityGroup, with the description Restrict inbound ports to HTTP and HTTPS. The trigger type was set to When configuration changes, and the scope was limited to AWS EC2 SecurityGroup resources. A parameter named debug with the value true was also added to enable additional debugging information from the Lambda function. After saving the rule, its details were examined, and the initial compliance evaluation was monitored until the resources were marked as Compliant.

Result: The custom AWS Config rule EC2SecurityGroup was created and configured to invoke the preconfigured Lambda function whenever a monitored EC2 security group changed. The initial evaluation identified the required security group configuration and reported the monitored resources as Compliant after remediation.

OUTPUT:

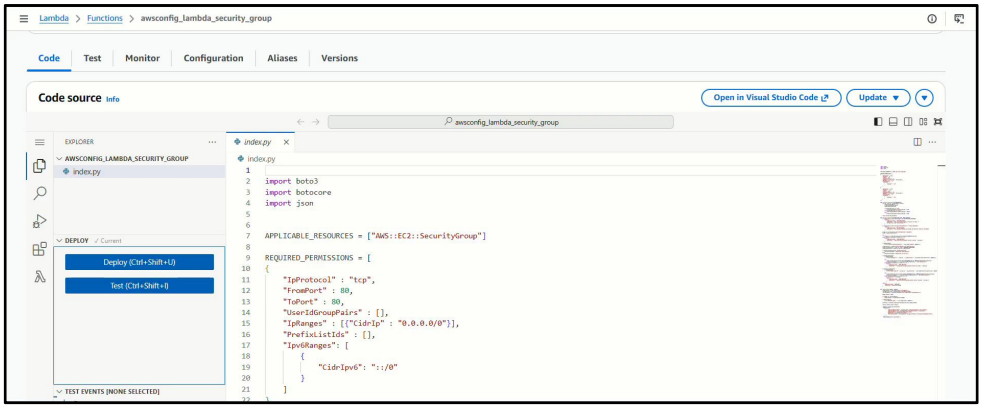


Task 5: Revisiting the Security Group Configuration

The Amazon VPC console was opened, and the LabSG1 security group in the Lab VPC was examined again. The inbound rules were reviewed to verify whether the additional SMTPS and IMAPS rules had been removed. It was observed that only HTTP and HTTPS traffic remained permitted. The HTTP and HTTPS rules were also configured for both IPv4 and IPv6, rather than only IPv4 as originally specified. The preconfigured Lambda function named awsconfig_lambda_security_group was then examined in the AWS Lambda console. Its Python code was reviewed to understand how it identifies the required inbound permissions, compares them with the existing security group configuration, and removes unauthorized rules. The debug parameter configured in the AWS Config rule was also examined.

Result: The security group configuration showed that the additional SMTPS and IMAPS inbound rules had been automatically removed. The Lambda function code confirmed the remediation logic used to restore the required security group configuration.

OUTPUT:



Task 6: Using CloudWatch Logs for Verification

The Amazon CloudWatch console was opened, and the Log groups section was accessed to locate the awsconfig_lambda_security_group log group. The Search all option was selected, and the log events were filtered using the search term `revoking for`. The matching log entries were expanded and examined to identify the actions performed by the Lambda function during remediation. The logs provided evidence that the additional SMTPS (TCP port 465) and IMAPS (TCP port 993) inbound rules had been revoked from the LabSG1 security group. Other log entries showing remediation actions on the remaining monitored security groups were also reviewed.

Result: CloudWatch logs confirmed that AWS Config invoked the Lambda function and that the function automatically removed the unwanted inbound rules from the monitored security groups.

OUTPUT:



BADGE:



PRESENTATION LINK:

[Insert YouTube / presentation link]

RESULT:

The lab demonstrated automated security monitoring and remediation using AWS Config and AWS Lambda. A custom Config rule was created to monitor EC2 security group changes, and additional inbound rules were introduced to simulate a security incident. The Lambda function automatically removed the unwanted SMTPS and IMAPS rules from the LabSG1 security group. CloudWatch logs were examined to verify the remediation actions and confirm that the unwanted rules had been revoked.